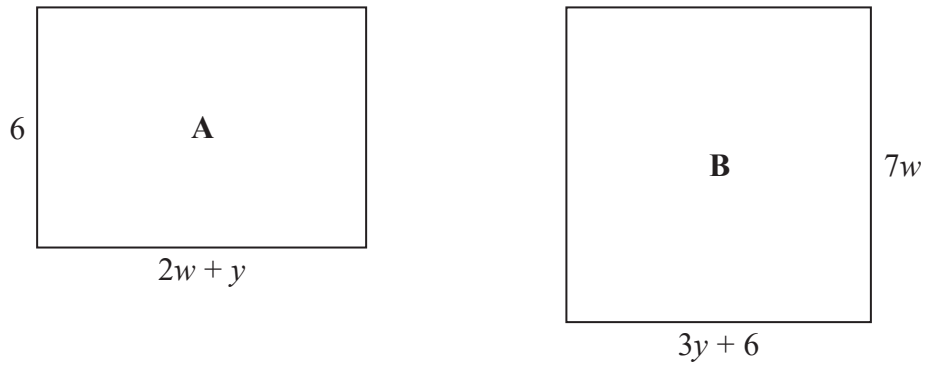
11 The diagram shows two rectangles, **A** and **B**.



All measurements are in centimetres.

The area of rectangle **A** is equal to the area of rectangle **B**.

Find an expression for y in terms of w.

Equal areas: 6(2w + y) = 7w(3y + 6)

12w + 6y = 21wy + 42w

6y – 21wy = 30w

y(6 – 21w) = 30w

y = 30w/(6 – 21w)

1 mark
form equal-area equation

1 mark
expand and collect

1 mark
factorise out y

1 mark
correct expression

30w/(6–21w)

(Total for Question 11 is 4 marks)

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